DANIEL SAM PETE THIYAGU

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Professional Experience

Senior Software Engineer, Media Understanding Team, Meta (07/2025 - Present)

- End-to-End Project Ownership: Led the technical design, infrastructure, and successful US launch of "Tune Your Algorithm" (TYA) on Instagram in under six months, taking the feature from inception to production.
- Massive User Engagement: Accelerated product adoption to 1.1% of all US Reels DAU within weeks of launch, driving a 30% week-over-week increase in daily TYA reel impressions.
- Launched TYA on Facebook to enable user agency and control for steering recommendations.
- Designed Agentic Systems with harness for user understanding to enable SOTA personalization
- Next-Gen Prototyping: Developed technical blueprint for Incremental Bio to save capacity of 65% GPUs saving $3M/yr
- Enabled various SOTA optimizations for various GenAI applications on User Understanding saving $2M/yr

Machine Learning Engineer, Core-ML Team, Google (06/2023-07/2025)

- Led the design and development of **LLM-powered Prompt Optimization** systems used by 20+ product areas, delivering a **30% performance improvement** via AI-driven tuning, rewriting, and re-ranking strategies.Extended prompt optimization to **multimodal inputs** including images and video, supporting a diverse range of GenAI tasks.
- Built tools for **AI-powered metaprompting** to automate prompt generation and rewriting, accelerating workflows and enabling **synthetic example generation** for enhanced few-shot prompting.
- Launched **Prompt Iterative Developer Journeys** and **Prompt Execution APIs**, powering LLM automation pipelines and integration in Search Review workflows.
- Applied advanced techniques such as **Chain-of-Thought**, **Self-Consistency**, and **Few-Shot Prompting**, alongside **benchmarking frameworks** to evaluate LLM performance and regressions.
- Founding engineer of **PromptLab**, prototyping AI agents with tool invocation, reasoning-based prompt chaining, and a scalable **Prompt Gallery** for prompt reuse and experimentation.
- Enabled GenAI-powered features for the **AI Content Safety** team, reducing review effort and improving takedown efficiency.
- Spearheaded infrastructure for **Prompt Execution Storage**, and led production efforts in **Security, Privacy, and Compliance** for enterprise LLM deployments.

Software Development Engineer, Core Team Google, Sunnyvale (03/2020-05/2023)

- Designed and deployed an **A/B classifier evaluation framework**, supporting model calibration and rollout for image and video content moderation systems.
- Integrated algorithms like **PhotoDNA**, fuzzy matching, and hash-based detection to boost system accuracy and scalability for takedowns.
- Owned and operated the **AI + Human hybrid review pipeline** for abusive content in image and video search, reducing takedown times from days to minutes. Improved automation and reduced redundancy in human reviews, saving **10% in operational resources**

Awards and Achievements

- Won 25+ Peer awards & Spot cash awards for excellent work in Google & Intuit
- Won **Core Impact Award at Google** along with few others for actionable reporting to NCMEC in Child Safety.

Education

- University of Massachusetts Amherst, Master of Science, Computer Science**,** GPA of **3.9/4.0** (09/2016 – 05/2018)
- CEG, Anna University, Bachelor of Engineering, Computer Science & Eng, CGPA of **9.39/10.0** (07/2010 – 05/2014)